

SEMINAR 6

1. Scrieti polinomul Taylor de grad 3
pt $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = e^{x+2y}$ asociat
pt f în $(0,0)$.

SOL:

$$\begin{aligned} T_3((x,y), (0,0)) &= f(0,0) + \frac{1}{1!} \left[\frac{\partial f}{\partial x}(0,0)(x-0) \right. \\ &\quad \left. + \frac{\partial f}{\partial y}(0,0)(y-0) \right] + \frac{1}{2!} \left[\frac{\partial^2 f}{\partial x^2}(0,0)(x-0)^2 + \right. \\ &\quad \left. + 2 \cdot \frac{\partial^2 f}{\partial x \partial y}(0,0)(x-0)(y-0) + \frac{\partial^2 f}{\partial y^2}(0,0)(y-0)^2 \right] \\ &\quad + \frac{1}{3!} \left[\frac{\partial^3 f}{\partial x^3}(0,0)(x-0)^3 + 3 \frac{\partial^3 f}{\partial x^2 \partial y}(0,0)(x-0)^2(y-0) \right. \\ &\quad \left. + 3 \frac{\partial^3 f}{\partial x \partial y^2}(0,0)(x-0)(y-0)^2 + \frac{\partial^3 f}{\partial y^3}(0,0)(y-0)^3 \right] \end{aligned}$$

$\frac{\partial f}{\partial x} = e^{x+2y}$	$\frac{\partial^2 f}{\partial x^2} = e^{x+2y}$	$\frac{\partial^3 f}{\partial x^3} = e^{x+2y}$
$\frac{\partial f}{\partial y} = 2 \cdot e^{x+2y}$	$\frac{\partial^2 f}{\partial x \partial y} = 2 \cdot e^{x+2y}$	$\frac{\partial^3 f}{\partial x^2 \partial y} = 2 \cdot e^{x+2y}$
	$\frac{\partial^2 f}{\partial y^2} = 4 \cdot e^{x+2y}$	$\frac{\partial^3 f}{\partial x \partial y^2} = 4 \cdot e^{x+2y}$
		$\frac{\partial^3 f}{\partial y^3} = 8 \cdot e^{x+2y}$

$$\begin{aligned} T_3((x,y), (0,0)) &= 1 + (x-0) + 2(y-0) + \frac{1}{2!} [(x-0)^2 + \\ &\quad + 2 \cdot 2(x-0)(y-0) + 4(y-0)^2] + \\ &\quad + \frac{1}{3!} [(x-0)^3 + 3 \cdot 2(y-0)(x-0)^2 + 3 \cdot 4(y-0)^2(x-0) + \\ &\quad + 8(y-0)^3] = \\ &= 1 + x + 2y + \frac{(x-0)^2}{2} + 2xy + 2y^2 + \frac{x^3}{6} + \\ &\quad + yx^2 + 2y^2x + \frac{4}{3}y^3 = \\ &= \frac{x^3}{6} + \frac{4y^3}{3} + yx^2 + 2y^2x + \frac{x^2}{2} + 2y^2 + 2xy + \\ &\quad + x + 2y + 1 = \\ &= \boxed{1 + (x+2y) + \frac{1}{2}(x+2y)^2 + \frac{1}{6}(x+2y)^3} \\ &\quad T_3((x,y), (0,0)) \end{aligned}$$

EXTREMELE LIBERE (SIMPLE) LOCALE ALE UNEI FCT DATE PE $\mathbb{R}^2/\mathbb{R}^3$

$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \dots$
m.d. $f \in C^2(\mathbb{R}^2)$

1) Rez. internur:

$$\begin{cases} \frac{\partial f}{\partial x}(x, y) = 0 \\ \frac{\partial f}{\partial y}(x, y) = 0 \end{cases}$$

facta remanere

$\Rightarrow Sol. (x_i, y_i) = \text{pt. crit.}$

2) $H(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$

3) $H(x_i, y_i) = \begin{pmatrix} \Delta_1 & \Delta_2 \\ \Delta_2 & \Delta_3 \end{pmatrix}$

• $\Delta_1 + \Delta_2 + \Delta_3 > 0 \Rightarrow (x_i, y_i) = \text{MINIM LOCAL}$

• $\Delta_1 + \Delta_2 + \Delta_3 < 0 \Rightarrow (x_i, y_i) = \text{MAXIM LOCAL}$

• $\Delta_2 - \Delta_1 \Delta_3 < 0 \Rightarrow (x_i, y_i) = \text{NUE PCTE EXTREM (PCT SA)}$

• $\Delta_2 = 0 \Rightarrow$ nu putem stabili natura punctului
 $\Rightarrow$ Studiem natura pt. def.

Met 3 $d^2 f(x, y, z)(a, b, c)$

poz. def (minimum)

Met 4 - definitia

neg. def (maximum)

$f: \mathbb{R}^3 \rightarrow \mathbb{R}, f(x, y, z) = \dots$

1) Rez. externur:

$$\begin{cases} \frac{\partial f}{\partial x}(x, y, z) = 0 \\ \frac{\partial f}{\partial y}(x, y, z) = 0 \\ \frac{\partial f}{\partial z}(x, y, z) = 0 \end{cases}$$

$\Rightarrow Sol. (x_i, y_i, z_i) = \text{PCT CRIT.}$

2) $H(x, y, z) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial y \partial z} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix}$

3) $H(x_i, y_i, z_i) = \begin{pmatrix} \Delta_1 & \Delta_2 & \Delta_3 \\ \Delta_2 & \Delta_4 & \Delta_5 \\ \Delta_3 & \Delta_5 & \Delta_6 \end{pmatrix}$

• $\Delta_1 + \Delta_2 + \Delta_3 + \Delta_4 + \Delta_5 + \Delta_6 > 0 \Rightarrow \text{MINIM LOCAL}$

• $\Delta_1 + \Delta_2 + \Delta_3 + \Delta_4 + \Delta_5 + \Delta_6 < 0 \Rightarrow \text{MAXIM LOCAL}$

• $\Delta_2 - \Delta_1 \Delta_3 < 0 \Rightarrow \text{PUNCT SA}$

• $\Delta_2 = 0 \Rightarrow ?$
cu ajuto

Met 2

200. pașaport

$\det(H(x_i, y_i, z_i) - \lambda I_3) = 0$

• $\lambda_1 + \lambda_2 + \lambda_3 > 0 \Rightarrow \text{MINIM}$

• $\lambda_1 + \lambda_2 + \lambda_3 < 0 \Rightarrow \text{MAXIM}$

• $\lambda_2 - \lambda_1 \lambda_3 < 0 \Rightarrow \text{PUNCT SA}$

• Altfel $\rightarrow ?$

1.

Eu

Extremwerte fct. $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = xy(x^2+y^2-4)$

SOL:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} 3x^2y + y^3 - 4y = 0 \\ x^3 + 3xy^2 - 4x = 0 \end{cases} \Rightarrow \begin{cases} y(3x^2 + y^2 - 4) = 0 \\ x(x^2 + 3y^2 - 4) = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ 3x^2 + y^2 - 4 = 0 \end{cases}$$

① $y = 0 \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = 0 \vee x = \pm 2$
 $\Rightarrow (0,0), (2,0), (-2,0)$

② $3x^2 + y^2 - 4 = 0 \Rightarrow y^2 = 4 - 3x^2 \Rightarrow$
 $\Rightarrow x(x^2 + 12 - 9x^2 - 4) = 0 \Rightarrow$
 $\Rightarrow x(-8x^2 + 8) = 0 \Rightarrow$
 $\Rightarrow x(1 - x^2) = 0 \Rightarrow$
 $\Rightarrow x = 0 \vee x = \pm 1$

$x = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2 \Rightarrow (0,2), (0,-2)$

$x = 1 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1 \Rightarrow (1,1), (1,-1)$

$x = -1 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1 \Rightarrow (-1,1), (-1,-1)$

$H(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 6xy & 3x^2 + 3y^2 - 4 \\ 3x^2 + 3y^2 - 4 & 6xy \end{pmatrix}$

$H(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \Delta_1 = 0, \Delta_2 = -16 < 0 \Rightarrow (0,0) \text{ pnt sa}$

$H(2,0) = \begin{pmatrix} 0 & 8 \\ 8 & 0 \end{pmatrix} \Rightarrow \Delta_1 = 0, \Delta_2 = -64 < 0 \Rightarrow (2,0) \text{ pnt sa}$

$H(1,1) = \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix} \Rightarrow \Delta_1 = 8 > 0, \Delta_2 = 32 > 0 \Rightarrow (1,1) \text{ pnt minimum local}$

2. Pot. de extrem $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x,y) = xye^{-(x^2+y^2)}$

Sol:

$\mathbb{R}^2$ - m.d. $f \in C^2(\mathbb{R}^2)$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} y(e^{-(x^2+y^2)} - 2x \cdot e^{-(x^2+y^2)}) = 0 \\ x(e^{-(x^2+y^2)} - 2y \cdot e^{-(x^2+y^2)}) = 0 \end{cases} \Rightarrow \begin{cases} y(1-2x) = 0 \\ x(1-2y) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} y \cdot e^{-(x^2+y^2)} (1-2x) = 0 \\ x \cdot e^{-(x^2+y^2)} (1-2y) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} y(1-2x) = 0 \\ x(1-2y) = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ x = 0 \end{cases} \Rightarrow x = 0 \vee y = 0$$

(1) $y = 0 \Rightarrow x = 0 \Rightarrow (0,0)$

(2) $x = 0 \Rightarrow y(1-2 \cdot 0) = 0 \Rightarrow 1-2y = 0 \Rightarrow y = \frac{1}{2}$

$y = \frac{1}{2} \Rightarrow x(1-2 \cdot \frac{1}{2}) = 0 \Rightarrow x = 0 \Rightarrow (0, \frac{1}{2})$

$x = \frac{1}{2} \Rightarrow y(1-2 \cdot \frac{1}{2}) = 0 \Rightarrow y = 0 \Rightarrow (\frac{1}{2}, 0)$

$$H(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} [e^{-(x^2+y^2)} - 2x \cdot e^{-(x^2+y^2)}] =$$

$$= -2xy \cdot e^{-(x^2+y^2)} (1-2x+y) =$$

$$= -2xy \cdot e^{-(x^2+y^2)} (1-2x+y)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} [e^{-(x^2+y^2)} - 2x \cdot e^{-(x^2+y^2)}] =$$

$$H(x,y) = \begin{pmatrix} -2xy(1-2x+y)e^{-(x^2+y^2)} & (1-2x)(1-2y)e^{-(x^2+y^2)} \\ (1-2x)(1-2y)e^{-(x^2+y^2)} & -2xy(1-2x+y)e^{-(x^2+y^2)} \end{pmatrix}$$

• $H(0,0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow \Delta_1 = 0, \Delta_2 = -1 < 0 \Rightarrow$ PUNCT SA

• $H(\frac{1}{2}, \frac{1}{2}) = \begin{pmatrix} -2e^{-1} & 0 \\ 0 & -2e^{-1} \end{pmatrix} \Rightarrow \Delta_1 = -2e^{-1} < 0, \Delta_2 = 4e^{-1} > 0 \Rightarrow$ PUNCT MAXIM

3. $f: (\mathbb{R}^+)^2 \rightarrow \mathbb{R}, f(x,y) = xy + \frac{y}{x^2} + \frac{x}{y^2}$

SOL:

$(\mathbb{R}^+)^2$ - m.d. $f \in C^2(\mathbb{R}^2)$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} y - \frac{y}{x^3} = 0 \\ x - \frac{x}{y^3} = 0 \end{cases} \Rightarrow \begin{cases} y(1 - \frac{1}{x^3}) = 0 \\ x(1 - \frac{1}{y^3}) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x - \frac{y}{x^3} = 0 \\ y - \frac{x}{y^3} = 0 \end{cases} \Rightarrow \begin{cases} x^4 - y = 0 \\ y^4 - x = 0 \end{cases} \Rightarrow \begin{cases} x^4 = y \\ y^4 = x \end{cases}$$

$$\Rightarrow x - \frac{x^4}{x^3} = 0 \Rightarrow x(1 - x) = 0 \Rightarrow \begin{cases} x = 0 \\ x = 1 \end{cases}$$

$x = 1 \Rightarrow y = 1 \Rightarrow (1,1)$ pt critic

$$H(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} -\frac{2y}{x^4} & 1 - \frac{1}{x^3} \\ 1 - \frac{1}{y^3} & -\frac{2x}{y^4} \end{pmatrix}$$

$$\Rightarrow H(1,1) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \Rightarrow \Delta_1 = -2 < 0, \Delta_2 = 4 > 0 \Rightarrow (1,1) \text{ Minimum Local}$$

4. $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$f(x,y) = x^4 + y^4$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} 4x^3 = 0 \\ 4y^3 = 0 \end{cases} \Rightarrow x = y = 0$$

$$H(x,y) = \begin{pmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{pmatrix}$$

$$\Rightarrow H(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\Rightarrow \Delta_1 = 0, \Delta_2 = 0 \Rightarrow$ cu pu pattern
nu se poate determina

Studiu mat. pt (0,0)
cu def.

• (a,b) pt de max local
de f $\forall \epsilon > 0$ $\exists \delta > 0$ de
f(x,y) $\leq f(a,b) + \epsilon, \forall (x,y) \in U$
(min local)

$f(x,y) = x^3 + y^3$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} 3x^2 = 0 \\ 3y^2 = 0 \end{cases} \Rightarrow x = y = 0$$

$$H(x,y) = \begin{pmatrix} 6x & 0 \\ 0 & 6y \end{pmatrix}$$

$$H(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\Delta_1 = 0, \Delta_2 = 0 \Rightarrow$ nu se poate determina

Studiu mat. pt
cu def.

$f(0,0) = 0$

Fie $V \in U(0,0) \Rightarrow$
 $\exists \epsilon > 0$ $\exists \delta > 0$ de

(a, b) nu e pt de extrem
 fcaos pt f .
 $\forall v \in \mathcal{D}(a, b), \exists (x_1, y_1) \in V,$
 $(x_2, y_2) \in V$ at
 $f(x_1, y_1) > f(a, b)$
 $f(x_2, y_2) < f(a, b)$

$$\begin{aligned}
 P \pm \left(\frac{\varepsilon}{2}, \frac{\varepsilon}{2} \right) &\Rightarrow \\
 \Rightarrow f\left(\frac{\varepsilon}{2}, \frac{\varepsilon}{2}\right) &= \frac{\varepsilon^3}{8} + \frac{\varepsilon^3}{8} + 1 > 1 \\
 P \pm \left(-\frac{\varepsilon}{2}, 0 \right) &\Rightarrow \\
 \Rightarrow f\left(-\frac{\varepsilon}{2}, 0\right) &= -\frac{\varepsilon^3}{8} + 0 + 1 < 1 \\
 \Rightarrow (0, 0) \text{ nu e pt de} & \\
 \text{extrem pt } f &
 \end{aligned}$$

$f(0, 0) = 0;$
 Eia, $\forall v \in \mathcal{D}(0, 0)$ avem
 $f(x, y) = x^4 + y^4 \geq 0 = f(0, 0)$
 $\forall (x, y) \in V$
 $\Rightarrow (0, 0)$ pt de minim
 (absolut)

Analog $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $f(x, y) = 3x^3 + y^4$
 $f(x, y) = x^4 - y^4$
 $= -x^3 - y^3 + 2$
 $= -x^4 - y^4 + 5$
 $= -x^3 + y^4 + 1$
 etc.

• $f(x, y) = -x^3 + y^4 + 1$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} -3x^2 = 0 \\ 4y^3 = 0 \end{cases} \Rightarrow x = y = 0 \Rightarrow (0, 0)$$

$$H(x, y) = \begin{pmatrix} -6x & 0 \\ 0 & 12y \end{pmatrix} \Rightarrow H(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\Rightarrow \Delta_1 = \Delta_2 = 0 \Rightarrow ?$

Studiem cu def. natura lui $(0, 0)$

$f(0, 0) = 1$

Fie $V \in \mathcal{D}(0, 0) : V = (-\varepsilon, \varepsilon) \times (-\varepsilon, \varepsilon), \varepsilon > 0$

Cum:

$$f\left(\frac{\varepsilon}{2}, \frac{\varepsilon}{2}\right) = \frac{\varepsilon^4}{16} + 1 > 1, f\left(\frac{\varepsilon}{2}, 0\right) = -\frac{\varepsilon^3}{8} + 1 < 1$$

$\Rightarrow (0, 0)$ nu e pt de ext.

5. $Q: \mathbb{R}^2 \rightarrow \mathbb{R}, Q(x, y) = x^3 + y^3 - 3x^2 + 6y^2 + 3x + 12y$

SOL:

$\mathbb{R}^2$ -m.d. $Q \in \mathcal{C}^2(\mathbb{R}^2)$

$$\begin{cases} \frac{\partial Q}{\partial x} = 0 \\ \frac{\partial Q}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} 3x^2 - 6x + 3 = 0 \\ 3y^2 + 12y + 12 = 0 \end{cases} \Rightarrow \begin{cases} x^2 - 2x + 1 = 0 \\ y^2 + 4y + 4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} (x-1)^2 = 0 \\ (y+2)^2 = 0 \end{cases} \Rightarrow (1, -2)$$

$$H(x, y) = \begin{pmatrix} 6x-6 & 0 \\ 0 & 6y+12 \end{pmatrix}$$

$$H(1, -2) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Delta_2 = 0 \Rightarrow ??$$

Studium natura der $(0,0)$ redef.

- $Q(1, -2) = -6, Q(x, y) = (x^3 - 3x^2 + 3x) + (y^3 + 6y^2 + 12y) + 1$
- Für $V \in \mathcal{U}(1, -2) \Rightarrow$
 $= (x-1)^3 + (y+2)^3$
 $\Rightarrow V = (1-\varepsilon, 1+\varepsilon) \times (-2-\varepsilon, -2+\varepsilon), \varepsilon > 0$

$$Q(x, y) = \varepsilon$$

$$Q\left(1+\frac{\varepsilon}{2}, -2\right) = \left(\frac{\varepsilon}{2}\right)^3 + 0 - 6 > -6$$

$$Q\left(1-\frac{\varepsilon}{2}, -2\right) = \left(-\frac{\varepsilon}{2}\right)^3 + 0 - 6 < -6$$

$\Rightarrow (1, -2)$ nur
punkt de
extrem

6.

Für $Q: \mathbb{R}^3 \rightarrow \mathbb{R}, Q(x, y, z) = x^2 + y^2 + z^2 + xz + yz - x - y - z + 1$

SOL: $\mathbb{R}^3$ -m.d. $Q \in \mathcal{C}^2(\mathbb{R}^3)$

$$\begin{cases} \frac{\partial Q}{\partial x} = 0 \\ \frac{\partial Q}{\partial y} = 0 \\ \frac{\partial Q}{\partial z} = 0 \end{cases} \Rightarrow \begin{cases} 2x + z - 1 = 0 \\ 2y + z - 1 = 0 \\ 2z + x + y - 2 = 0 \end{cases}$$

$$\Rightarrow 4x + 4y + 4z - 4 = 0 \Rightarrow$$

$$\Rightarrow x + y + z = 1 \Rightarrow z = 0$$

$$\Rightarrow 2y + z - 1 = 0$$

$$2y + 4 - 2 = 0$$

$\Rightarrow (0,0,1)$ - point critique

$$H(x,y,z) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

$$\Delta_1 = 2 > 0$$

$$\Delta_2 = 3 > 0$$

$$\Delta_3 = 8 + 2 - 2 - 2 - 2 = 4 > 0$$

Méthode 2 (par parrain)

car. ec. $\det(H(0,0,1) - \lambda I_3) = 0$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow$$

$$\Rightarrow (2-\lambda)^3 + 2 - 3(2-\lambda) = 0 \quad 2-\lambda = t \Rightarrow$$

$$\Rightarrow t^3 - 3t + 2 = 0 \Rightarrow t_1 = 1$$

$$\Rightarrow (t-1)(t^2+t-2) = 0$$

$\hookrightarrow 1$
 $\hookrightarrow -2$

$$\begin{cases} \lambda_1 = \lambda_2 = 1 > 0 \\ 2 - \lambda = 2 \Rightarrow \lambda_3 = 4 > 0 \end{cases} \Rightarrow (0,0,1) \text{ - POINT DE MINIM LOCAL}$$

effet 3

$$\begin{aligned} d_x^2 f(0,0,1)(a,b,c) &= \frac{\partial^2 f}{\partial x^2}(0,0,1) a^2 + \frac{\partial^2 f}{\partial x \partial y}(0,0,1) b^2 + \frac{\partial^2 f}{\partial x \partial z}(0,0,1) c^2 + \\ &+ 2 \cdot \frac{\partial^2 f}{\partial x \partial y}(0,0,1) ab + 2 \cdot \frac{\partial^2 f}{\partial x \partial z}(0,0,1) ac + \\ &+ 2 \frac{\partial^2 f}{\partial y \partial z}(0,0,1) bc = 2a^2 + 2b^2 + 2c^2 + 2ab + \\ &\quad + 2bc + 2ac \end{aligned}$$

$$\begin{aligned} d_x^2 f(0,0,1)(a,b,c) &= 2(a^2 + ab + ac) + 2ab^2 + 2b^2 + 2bc \\ &= 2(a^2 + ab + ac + \frac{b^2}{2} + \frac{c^2}{2} + \frac{bc}{2}) - \frac{b^2}{2} - \frac{c^2}{2} - \\ &\quad - bc + 2b^2 + 2c^2 + 2bc = \\ &= 2(a + \frac{b}{2} + \frac{c}{2})^2 + 3\frac{b^2}{2} + 3\frac{c^2}{2} + bc = \end{aligned}$$

$$\begin{aligned}
 &= 2 \left(\dots \frac{1}{2} + \frac{1}{2} \left(b^2 + \frac{1}{2} bc \right) + \frac{1}{2} x^2 \right) \\
 &= 2 \left(\dots \frac{1}{2} + \frac{1}{2} \left(b^2 + \frac{1}{2} bc + \frac{1}{2} x^2 \right) - \frac{1}{2} \right) \\
 &= 2 \left(\dots \frac{1}{2} + \frac{1}{2} \left(b + \frac{1}{2} \right) + \frac{1}{2} x^2 \right) \geq 0, \forall a, b, c \in \mathbb{R} \\
 &\Rightarrow \text{pos. def} \Rightarrow (0, 0, 1) \text{ pos. minimum}
 \end{aligned}$$

7. $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \ln(1 + |x - y|)$

8. $f: \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}$

$$f(x, y) = xy \ln(x^2 + y^2)$$

Sol:

$$\frac{\partial f}{\partial x} = 0$$

$$\frac{\partial f}{\partial y} = 0$$